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Project Overview

Learning to play a musical instrument requires significant upfront investment. Beginners are hesitant to purchase expensive instruments before knowing if they'll stick with it, and music schools struggle to manage their rental inventory efficiently. A rental platform can make music education more accessible while helping rental providers manage their inventory and bookings.

Build a Music Instrument Rental Platform where users can browse available instruments, request rentals for specific time periods, and manage their bookings. The platform should include an admin panel for managing instruments, rental requests, and users.

You must:

1. **Gather requirements** - What does an instrument rental platform need? What features are essential?
 2. **Create user personas** - Who will use this system? What are their goals?
 3. **Design wireframes** - How should the instrument listings look? What's the rental request flow?
 4. **Build an ERD** - What database tables do you need? How do they relate?
 5. **Write user stories** with acceptance criteria
 6. **Plan your sprints** - Break the work into manageable tasks using a Trello/Jira board
 7. **Build the application** - Implement your design using PHP or Symfony
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Core Features (Must Have)

These features **must** be implemented for the project to be considered complete:

Customer Features:

- User registration and login
- Browse available instruments (guitar, piano, violin, drums, etc.)

- View individual instrument details (name, category, daily rental price, condition, description, image)
- Filter instruments by category (strings, percussion, woodwind, brass, keyboard)
- Search instruments by name or category
- Submit rental request (select instrument, start date, end date, purpose/reason for rental)
- View own rental requests and their status (pending, approved, rejected, completed)
- Cancel pending rental requests
- Update/edit profile information

Admin Features:

- Admin authentication (admin role in user table)
- Instrument management (create, edit, delete instruments)
- Instrument image uploads
- Category management (create, edit, delete categories)
- View all rental requests
- Approve or reject rental requests
- Mark rentals as "returned" when completed
- User management (view all users, block/unblock accounts)
- Track instrument availability (instruments marked as unavailable during rental period)

General Requirements:

- Responsive design (mobile and desktop)
- Form validation
- Date validation (end date must be after start date, no past dates)
- Secure authentication (hashed passwords, session management)
- Authorization checks (users can only see their own rental requests, admins see all)

Nice to Have Features

If you complete the core features and have time, consider adding **up to 2** of these:

- Calculate total rental cost automatically (daily rate × number of days)
- Rental history for users (see past completed rentals)
- Instrument ratings and reviews (after rental is completed)
- Featured/popular instruments section on homepage
- Calendar view showing instrument availability

- Email notifications for rental approval/rejection

Important: Do not attempt nice-to-have features until core features are fully working.

Technical Requirements:

- **Backend:** PHP or Symfony (team decides)
 - **Frontend:** HTML, CSS, JavaScript (Bootstrap, libraries are allowed)
 - **Database:** MySQL
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Requirements Engineering & Agile Deliverables

Before you start coding, you must produce:

1. **User Stories**
 - **Format:** "As a [user type], I want to [action], so that [benefit]"
 - **Example:** "As a user, I want to request a guitar rental for 2 weeks, so I can try learning before buying"
 2. **Acceptance Criteria**
 - Define what "done" means for each user story
 - Example: "Rental request form includes instrument selection, start date, end date, and rental purpose"
 3. **Entity-Relationship Diagram (ERD)**
 - Show all database tables and their relationships
 - Include primary keys, foreign keys, and important fields
 4. **Wireframes**
 - Sketch key pages: instrument list, instrument detail, rental request form, my rentals, admin dashboard
 - Can be hand-drawn or digital (Figma, Balsamiq, etc.)
 5. **Sprint Plan**
 - Break your project into 2 sprints (~1 week each)
 - Define what you'll complete in each sprint
 6. **Task Board**
 - Use Trello, Jira, or similar tool
 - Track tasks (Minimum: Backlog → UpNext → InProgress → Done)
 - Backlog needs to be finished before Project Start
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Research & Inspiration Sources

Before designing and coding, research how real rental platforms work to inform your decisions.

Rental/Booking Sites to Study

Browse these for user flows and UI patterns:

- **Airbnb** - <https://airbnb.com/> - Booking flow, calendar selection, listing details
 - **Rent the Runway** - <https://renttherunway.com/> - Rental marketplace, request system
 - **Fat Llama** - <https://fatllama.com/> - Equipment rental platform
 - **Sparetoolz** - <https://sparetoolz.com/> - Tool rental platform
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Database Design

- **dbdiagram.io** - <https://dbdiagram.io/>
 - **Lucidchart** - <https://lucidchart.com/>
 - **Database Answers** - <https://databaseanswers.org/>
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Wireframe Tools

- **Excalidraw** - <https://excalidraw.com/>
 - **Figma** - <https://figma.com/>
 - **draw.io** - <https://draw.io/>
 - **Miro** - <https://miro.com/>
 - **Pen and paper** - Sometimes the fastest option!
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UI Component Inspiration

- **Start Bootstrap** - <https://startbootstrap.com/>
 - **Bootstrap Made** - <https://bootstrapmade.com/>
 - **Colorlib** - <https://colorlib.com/wp/free-bootstrap-templates/>
 - **ThemeWagon** - <https://themewagon.com/theme-tag/free/>
 - **Bootsnipp** - <https://bootsnipp.com/>
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Icon Libraries

- **Font Awesome** - <https://fontawesome.com/icons>
 - **Heroicons** - <https://heroicons.com/>
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Color & Design Tools

- **Coolors** - <https://coolors.co/>
- **Color Hunt** - <https://colorhunt.co/>
- **Adobe Color** - <https://color.adobe.com/>

Typography

Google Fonts - <https://fonts.google.com/>

Image Resources

- **Unsplash** - <https://unsplash.com/>
- **Pexels** - <https://pexels.com/>
- **Pixabay** - <https://pixabay.com/>
- **Lorem Ipsum** - <https://picsum.photos/>

Add submission

Submission status

Submission status	No submissions have been made yet
Grading status	Not graded
Last modified	-
Submission comments	> Comments (0)